

QUALITY ASSURANCE (QA) PLAN

List Of Contributors

Hatice Melike Aydin 231101063

İclal Zeynep Kaya 231101073

Anıl Özişler 211101081

Task Matrix

Hatice Melike Aydin	İclal Zeynep Kaya	Anıl Özişler
Quality Assurance Strategy	Quality Factors & Metrics	Test Plan

Table Of Contents

1. Quality Assurance Strategy
 - a. Overview
 - b. Testing Methodologies
 - c. Automated vs. Manual Testing
2. Quality Factors & Metrics
3. Test Plan
 - a. Test Cases
 - b. Bug Tracking

1. Quality Assurance Strategy

Overview: Quality Assurance and Testing processes of the project will be systematically conducted through methodologies of testing to ensure the reliability, secureness and functionality of the product. Testing will start with the unit test to ensure that the components secure logical integrity. As the following step, integrity test will be conducted to test the harmony between modular components. As system tests, usability testing and performance testing will be conducted to ensure optimality of the product. A continuous testing procedure and bug tracking within the development process will be prioritized for the reliability of the final product.

Testing Methodologies

- **Unit Testing:** Jest for frontend and backend testing. Verifies the functionality and stability of the smallest components in isolation.
- **Integration Testing:** Confirms correct data flow and communication between distinct modular components.
- **System Testing:** Evaluates the application as a unified whole against its specified requirements.
- **Security Testing:** Identifies vulnerabilities and ensures data protection.
- **Acceptance Testing:** Confirms the final product aligns with user needs and the documented business objectives.
- **Regression Testing:** Ensures that new code changes do not introduce bugs in previously working functionality.

Automated vs. Manual Testing

Test Type	Automated	Manual
Unit Tests	Jest for frontend and backend testing	Not applicable
UI Testing	Selenium for automated UI validation	Manual user interface validation
Security Testing	Automated vulnerability scanning	Manual penetration testing
Regression Testing	Automated test suites to validate previous functionality	Manual verification for exceptional scenarios
Acceptance Testing	Non-executable	Final validation based on business requirements

2. Quality Factors and Metrics

Quality Factor	Description	Measurement Metric
Performance	Response time	< 5 seconds
Security	Secure authentication	No security issues expected
Usability	User-friendliness	80%+ satisfaction
Reliability	System uptime	85% uptime

3. Test Plan

Test Cases

Test Name	Description	Expected Outcome
User Authentication	User logs in or signs up with a username and password	System securely logs in or registers the user
Free Time Syncing	User searches for overlapping free time slots with another user	System successfully lists the right overlapping free time slots
Sync request	User sends sync request	System successfully syncs the users
Schedule enrollment	User enters their personal schedule	System enrolls the users schedule successfully without any missing information
Profile update	Users update their account information	System successfully updates users information without any missing or incorrect information

Bug Tracking

- GitHub issues will be the main bug tracking mechanism
- Bug report contents
 - Severity level
 - Actual outcome and Expectation
 - Reproduction steps